

Introduction To Algorithms

Week 3 - Data Structure Choice:
"Choosing How the Data Lives"

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Terms to Remember

Term	Brief Meaning
Data structure	A way to organize data for use by an algorithm
Access pattern	The operations the algorithm performs most often
List	Ordered, changeable collection
Tuple	Ordered, fixed collection
Dictionary	Key-value structure for direct lookup
Set	Collection focused on uniqueness and membership
Series	One-dimensional labeled data structure
Dataframe	Two-dimensional table-like data structure
Matrix	Rectangular arrangement of values used in mathematical operations
Abstract data type	A structure defined by behavior, not one specific implementation
Stack	Last-in, first-out access pattern (LIFO)
Queue	First-in, first-out access pattern (FIFO)
Tree	Hierarchical structure with parent-child relationships

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What We Will Use Today

- list, tuple, dictionary, and set behavior
- operation-based comparison
- access pattern as the reason for choosing a structure
- Series/Dataframe and matrix recognition
- stacks, queues, and trees as abstract patterns
- Lab 03 comparison table

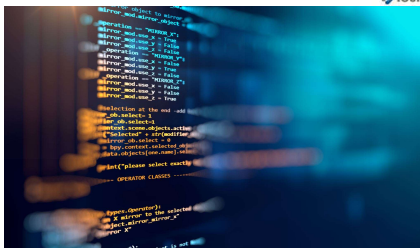
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What We Will "Touch On" and Revisit Later

Deeper search and sort behavior
Graph structures
Recursion and tree traversal
Matrix-heavy algorithms
Data modeling and analytics structures
Library-specific performance details



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Review Coding Lab

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Review: What Lab 02 Taught Us

Last week, we focused on:

input size
timing evidence
repeated work
growth pattern
limitation notes

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Today's Question

What structure makes this problem easier to solve?

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Data Structures Shape Algorithms

- How data is stored
- How data is found
- How data is updated
- How data is removed
- How the code reads

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Same Data, Different Structures, Different Algorithm Behavior

The same four student records stored in three common structures.

1. List

Ordered collection (by position)

0	Maya
1	Jordan
2	Luis
3	Asha

Access Pattern: Scan in order

0 → 1 → 2 → 3

2. Dictionary

Key-value pairs (lookup by key)

101	→	Maya
102	→	Jordan
103	→	Luis
104	→	Asha

Access Pattern: Lookup by key

103 → Luis

3. Set

Unordered collection (unique items)

Maya	Jordan
Luis	Asha

Access Pattern: Check membership

Luis → ✓

Structure changes the path the algorithm takes.

Data "Shapes" Behavior

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Familiar Is Not Always Fit

A familiar structure may work.
But a better-fitting structure may make the solution:

- Clearer
- Shorter
- Easier to test
- Easier to update
- More efficient



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Textbook Review

Four Main Areas

The reading covers:

1. Built-in data types
2. Series and DataFrames
3. Matrices and matrix operations
4. Abstract data types

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Textbook Review

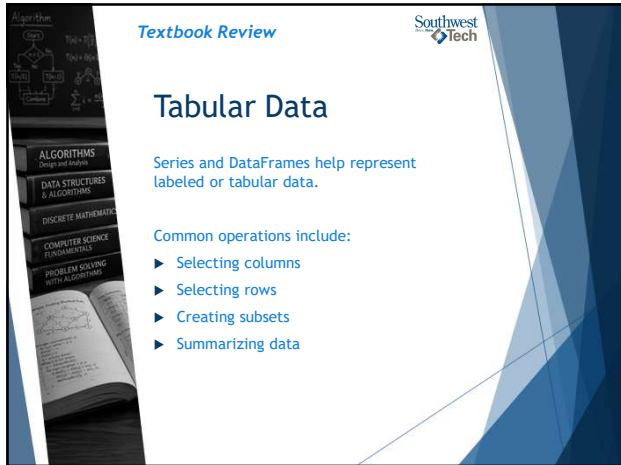
Built-In Types

Built-in structures include:

- List
- Tuple
- Dictionary
- Set

Each supports different operations.

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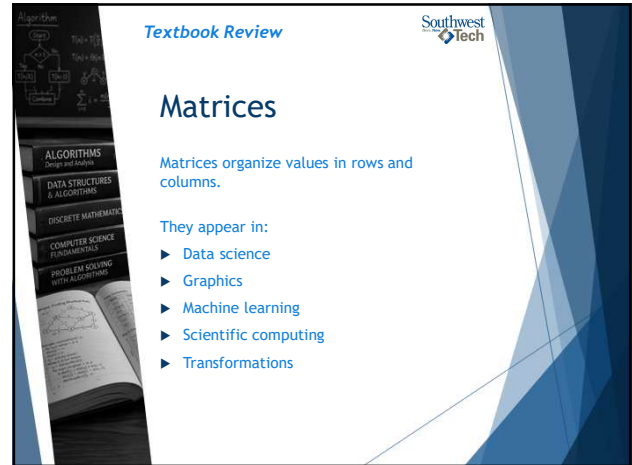
Tabular Data

Series and DataFrames help represent labeled or tabular data.

Common operations include:

- ▶ Selecting columns
- ▶ Selecting rows
- ▶ Creating subsets
- ▶ Summarizing data

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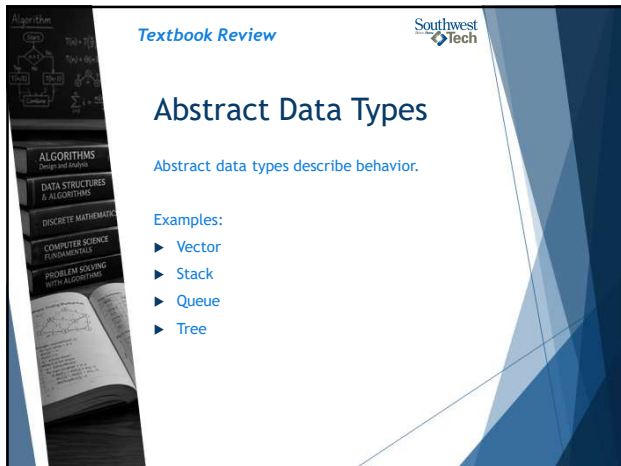
Matrices

Matrices organize values in rows and columns.

They appear in:

- ▶ Data science
- ▶ Graphics
- ▶ Machine learning
- ▶ Scientific computing
- ▶ Transformations

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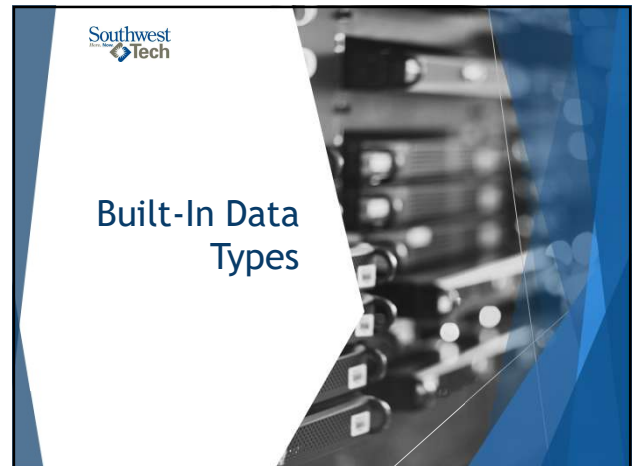
Abstract Data Types

Abstract data types describe behavior.

Examples:

- ▶ Vector
- ▶ Stack
- ▶ Queue
- ▶ Tree

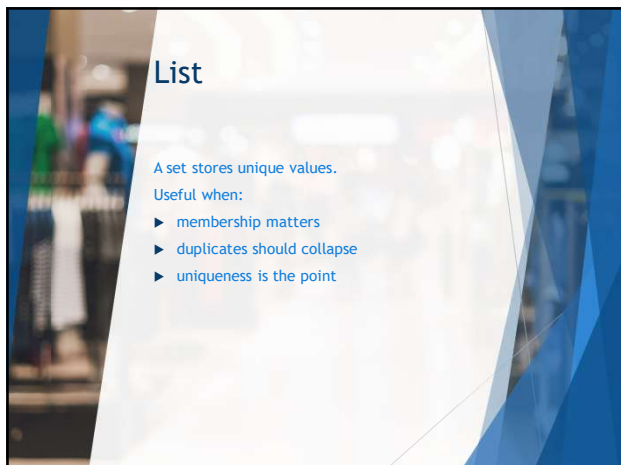
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Built-In Data Types

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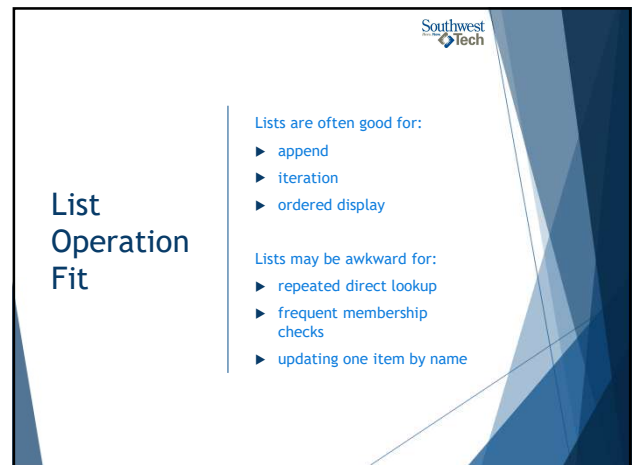


List

A set stores unique values.
Useful when:

- ▶ membership matters
- ▶ duplicates should collapse
- ▶ uniqueness is the point

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List Operation Fit

Lists are often good for:

- ▶ append
- ▶ iteration
- ▶ ordered display

Lists may be awkward for:

- ▶ repeated direct lookup
- ▶ frequent membership checks
- ▶ updating one item by name

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Tuple

A tuple is an ordered, fixed collection.

Useful when:

- ▶ values belong together
- ▶ the group should not change
- ▶ position has meaning

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Dictionary

A dictionary stores key-value pairs.

Useful when:

- ▶ each item has an identifier
- ▶ direct lookup matters
- ▶ updates are based on a key
- ▶ labels make code clearer

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Dictionary Operation Fit

Dictionaries often fit:

- ▶ lookup by key
- ▶ update by key
- ▶ counting
- ▶ grouping records

Watch for:

- ▶ key choice
- ▶ missing keys
- ▶ order requirements

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Set

A set stores unique values.

Useful when:

- ▶ membership matters
- ▶ duplicates should collapse
- ▶ uniqueness is the point

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Set Operation Fit

Sets often fit:

- ▶ membership checks
- ▶ duplicate removal
- ▶ uniqueness tracking
- ▶ intersections and differences

Sets may not fit:

- ▶ preserving duplicates
- ▶ preserving original order
- ▶ storing attached values

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Built-In Structure Comparison

Ask what the problem needs:

- ▶ order?
- ▶ fixed grouping?
- ▶ key-based lookup?
- ▶ uniqueness?
- ▶ duplicates?
- ▶ updates?

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Four Built-In Data Structures

1. List

Maya	Jordan	Luis	Asha
0	1	2	3

Ordered, changeable

2. Tuple

(Maya, Jordan, Luis, Asha)

Ordered, fixed

3. Dictionary

101	→	Maya
102	→	Jordan
103	→	Luis
104	→	Asha

Lookup by key

4. Set

Maya Jordan Luis Asha

Maya

Unique membership

Built-In DS's

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Complexity Tables Are Reference Tools

When you see Big-O operation tables:

- ▶ Do not panic
- ▶ Find the operation
- ▶ Compare likely costs
- ▶ Connect cost to your use case
- ▶ -Verify with evidence when needed

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Using Operation Tables as Your Reference Map

Operation	What It Does	Typical Use
lookup	Find a value by key or position.	Search for an item.
insert	Add a new item to the structure.	Add new data.
delete	Remove an item from the structure.	Remove old or unwanted data.
update	Change the value of an existing item.	Modify existing data.

Which structure fits this operation?

Match the operation to the best data structure for the job.

Use the table to guide choices, not as a memory test.

Textbook Op Table

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Operation Names Matter

Avoid

"Dictionaries are better"

Prefer

"For lookup by student name, the dictionary is a better fit."

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Operation Fit and Complexity

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Access Pattern

Access pattern means - How data is usually:

- ▶ Added (C)
- ▶ Found (R)
- ▶ Updated (U)
- ▶ Removed (D)
- ▶ Displayed (V)

(Think of CRUD-V)

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Operation	Structure A	Structure B	Better Fit	Why?
Lookup	List scan	Dictionary Key	Dictionary	Direct Lookup
...				

Operation Comparison Table

This is the evidence format you will use in this week's lab

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Choosing the Better Data Structure

Match the operation to the structure that works best.

Operation	Better Fit	Why
Find by ID	Dictionary	direct key lookup
Keep order	List	position matters
Remove duplicates	Set	unique membership

 The reason matters more than the label.

Op Comparison Table

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Complexity And Clarity

A good structure choice considers:

- ▶ Correctness
- ▶ Operation cost
- ▶ Code clarity
- ▶ Memory use
- ▶ Future change

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AI-Assisted Structure Review

AI can help review a structure choice.

Ask AI to:

- ▶ identify the access pattern
- ▶ compare two structures
- ▶ explain tradeoffs
- ▶ suggest an alternative
- ▶ avoid rewriting unless asked

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Useful AI Prompt Pattern

"I am solving this problem with [structure A] and [structure B].

Compare them for add, lookup, update, remove, and display.

Explain which structure fits the access pattern better.

Do not rewrite my code unless I ask."

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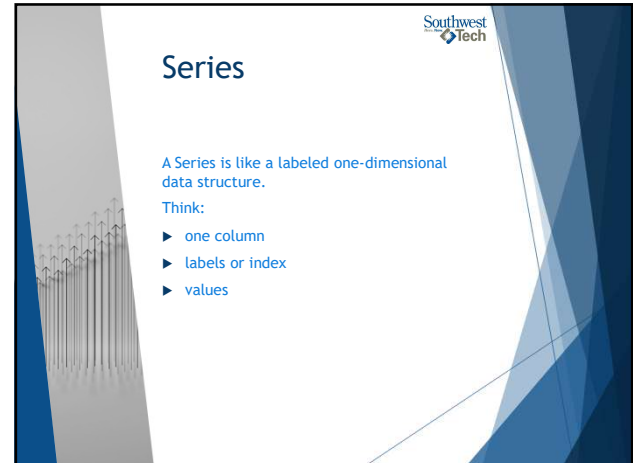
Break

10 Minute "Human" bio-break

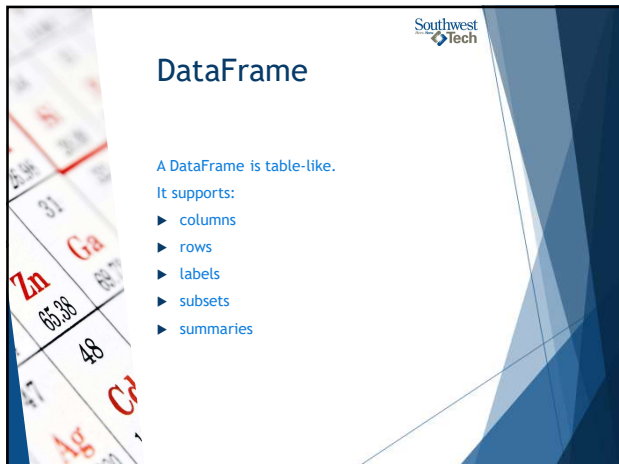
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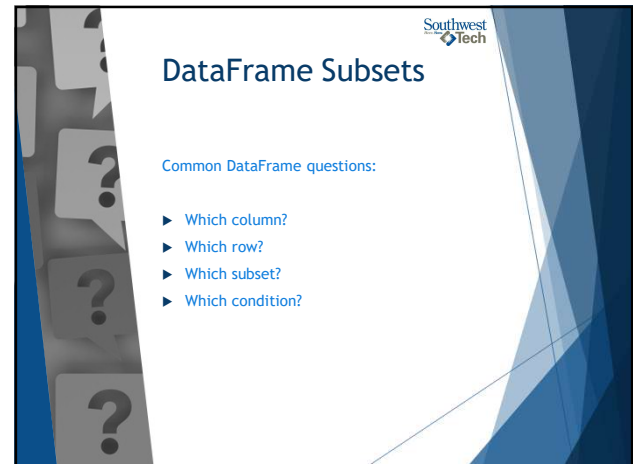
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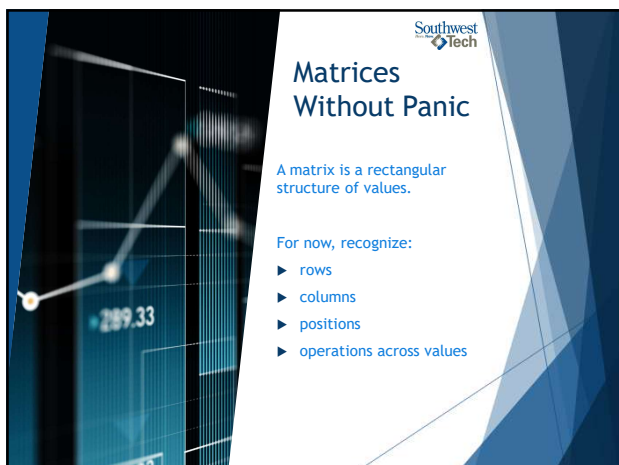
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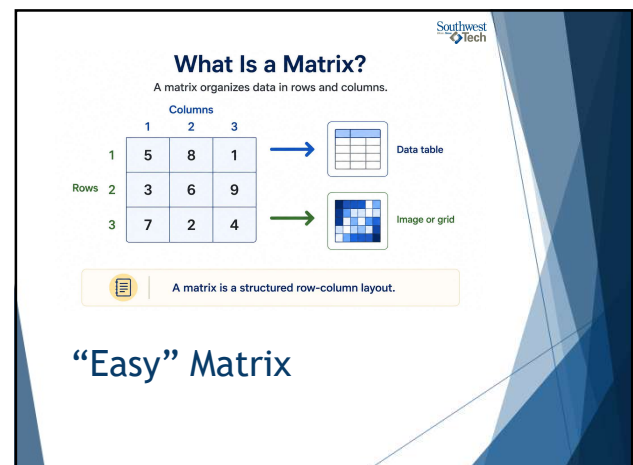
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Visualizing Matrices: ("Early" Use of Matrices)

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1D Matrix (Series)
One dimension
 $1 \times n$

2D Matrix (Square)
Two dimensions (rows $\times$ columns)
 $m \times n$

3D Matrix (Cube)
Three dimensions (depth $\times$ rows $\times$ columns)
 $p \times m \times n$

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THE MATRIX (MOVIE)

THE REAL WORLD vs. THE MATRIX

THE REAL WORLD
Bodies imprisoned. Mind unaware.

THE MATRIX
A simulated reality generated by machines.

3D MATRIX (NESTED MATRIX)
THE REAL WORLD vs. THEIR MAKE BELIEVE WORLD

THE REAL WORLD
The larger 3D matrix. People are trapped inside.

THEIR MAKE BELIEVE WORLD
A nested 3D matrix within the larger matrix. Their perceived reality.

Outer 3D Matrix = The Real World (Bodies imprisoned, true reality)
Inner 3D Matrix = Their Make Believe World (Simulated reality, generated as real)
People = Trapped, living in a simulation

People are trapped in the real world, living in a simulation created to control them.

Each person lives in their own nested simulation, unaware of the larger reality that contains them.

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Matrix Operations

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Matrix operations may involve:

- ▶ accessing a value
- ▶ adding matrices
- ▶ multiplying matrices
- ▶ transforming data

Complexity depends on the operation and dimensions.

$m = n$

$N = \sqrt{(x_2 - x_1)}$

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Data Structures For AI And Analytics

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Later AI and analytics work uses:

- ▶ tables
- ▶ vectors
- ▶ matrices
- ▶ graphs
- ▶ indexes

This week 3 gives you the recognition layer.

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What To Deeply Learn vs Recognize

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Deeply learn now:

- ▶ list
- ▶ dictionary
- ▶ set
- ▶ operation comparison

Recognize for now:

- ▶ Series/DataFrame
- ▶ matrix
- ▶ vector
- ▶ stack/queue/tree

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Abstract Data Types

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Abstract Data Type

An abstract data type is defined by behavior.

It asks:

- ▶ What operations are allowed?
- ▶ What access pattern is expected?
- ▶ What rules shape use?

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Abstract Data Types

Four Abstract Data Types: Different Access Pattern

1. Stack

last in, first out

2. Queue

first in, first out

3. Tree

parent-child paths

4. Vector

direction or ordered values

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Vectors:

A vector is an ordered collection of values.

It may represent:

- ▶ Coordinates
- ▶ Measurements
- ▶ Features
- ▶ Direction or magnitude



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Stack

A stack uses (LIFO) last-in, first-out behavior.

Think:

- ▶ undo history
- ▶ browser back stack
- ▶ nested tasks
- ▶ function call stack

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Queue

A queue uses (FIFO) first-in, first-out behavior.

Think:

- ▶ help desk tickets
- ▶ print jobs
- ▶ checkout line
- ▶ task scheduling



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Tree

A tree represents hierarchy.

Think:

- ▶ folders
- ▶ organization charts
- ▶ decision trees
- ▶ parent-child relationships

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Abstract Types By Access Pattern

Need	Last item first? → Stack.
Need	First item first? → Queue.
Need	Hierarchy? → Tree.
Need	Ordered values? → Vector.

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Week 3 Decision Frame

Choose a structure by asking:

- ▶ What data do I store?
- ▶ What operation happens most?
- ▶ What should be fast or clear?
- ▶ What tradeoff am I accepting?

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Demo

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Demo Scenario:

Problem:

- Compare:
- list of names
 - dictionary of attendance counts

Evidence:

1.?

Goal

Track attendance check-ins

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Demo Core Operation



Core operation:

Record one more check-in for Ava.



Then inspect:

How the list changes
How the dictionary changes
How lookup works

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Demo Evidence

Before/after representation

Lookup result

Operation comparison

Better fit

Why the fit is better

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Attendance Tracking: Two Data Structure Versions

Same goal. Different structure. Different access path.

List Version

Before

- Maya
- Jordan
- Luis

After

- Maya
- Jordan
- Luis
- Asha

Add Asha to attendance

Dictionary Version

Before

101	→	Maya
102	→	Jordan
103	→	Luis

After

101	→	Maya
102	→	Jordan
103	→	Luis
104	→	Asha

Add key 104 with value Asha

Same data, different lookup and update path. Choose the structure that fits the operation you need.

Demo Evidence

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Demo Explanation

Good explanation:

- ▶ Names the operation
- ▶ Compares both structures
- ▶ Identifies the better fit
- ▶ Explains the tradeoff

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Demo Add-On: Console Color



After the plain demo works, add light console color.

The color helps highlight:

- ▶ section headings
- ▶ passed tests
- ▶ failed tests
- ▶ important boundary cases

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From Demo to Lab



In Lab 03, choose a different scenario and compare two structures.

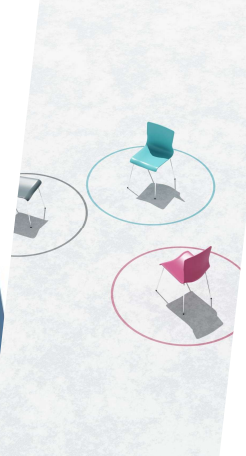
Your work must include:

- ▶ Two structures
- ▶ At least three operations
- ▶ Implementation or simulation
- ▶ Comparison table
- ▶ Recommendation

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Walkthrough Support



The walkthrough artifact can help you:

- ▶ Consider data types
- ▶ Identify operations
- ▶ Better fit
- ▶ Missing data

If you use it, you must explain the pattern in your own words, what you changes, and your decisions.

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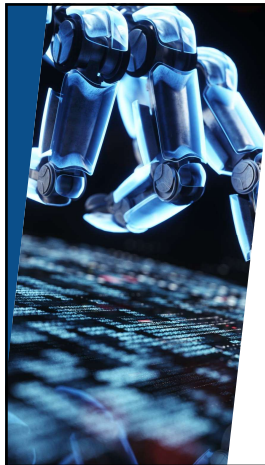
README Evidence



Your README should show:

- ▶ Problem Statement
- ▶ Structures compared
- ▶ Operations tested
- ▶ Comparison table
- ▶ Recommendation
- ▶ AI-use note, if applicable

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AI Use In Lab 03

Manual first:

- ▶ Choose two structures
- ▶ Explain your first comparison
- ▶ Identify key operations

Then AI may help:

- ▶ Suggest alternatives
- ▶ Critique tradeoffs
- ▶ Clarify operation fit

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Wrap up

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What To Carry Forward

Data structures are choices.

Choose based on:

- ▶ operations
- ▶ access patterns
- ▶ clarity
- ▶ tradeoffs

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Lab 03 Success Check

- ▶ Uses two structures meaningfully
- ▶ Compares specific operations
- ▶ Explains tradeoffs
- ▶ Avoids habit-only choices
- ▶ Gives a clear recommendation

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Next Step

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For Lab 03:

- Choose a scenario
- Choose two structures
- Compare at least three operations
- Build the comparison table
- Recommend the better fit

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How To Use The Textbook For Next Week's Reading

Next week focuses on search and sort behavior. As you read, focus on:

- ▶ Sorting as a way to arrange data
- ▶ Searching as a way to find a target
- ▶ Variable swapping as a small mechanical step
- ▶ Linear search, binary search, and interpolation search
- ▶ Why binary search requires sorted data
- ▶ Performance differences among search and sort strategies
- ▶ Practical lookup ideas such as primary and foreign keys



Treat Natural Language Processing (NLP) as context, not as a new coding target.

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Preparing for Next Week: Search and Sort

Before we choose an algorithm, check the setup.

1. Sorted List
Data is in order.
3 7 12 18 24 31

→

2. Search Target
What are we looking for?
Target = 18

→

3. Precondition Checklist

- ✓ sorted data
- ✓ clear target
- ✓ compare step
- ✓ trace evidence

 Some algorithms only work when their assumptions are true.

Reading Preview

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Thank you! Have Fun!

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